

BOEKBESPREKINGS / BOOK REVIEWS

Introduction to Probability and Statistics, Part II: Statistics

by Narayan C. Giri

First edition, 314 pages, Marcel Dekker, Inc., New York, 1975. \$14.75

The first of this two-part publication, reviewed in this journal (Volume 9, No. 1, 1975), was concerned with probability theory and probability distributions. The second part comprises seven chapters which cover statistical inference, tests of hypotheses, order statistics, sequential analysis, non-parametric methods, and the linear model of full rank. Each chapter is concluded with a set of exercises, while a standard set of statistical tables is published in an appendix. This text features the modern prominence of matrix algebra in texts for introductory courses and accordingly a relatively large appendix of standard algebra of matrices is included.

Although the contents of this text are presented in a clear and logical manner, it is difficult to recommend it for use as a teaching reference. In my opinion the general level is too advanced for a first course in theoretical statistics. There is little motivation for the methods used, and with the exception of isolated cases, there are no worked examples. However, since the chapters are virtually unrelated presentations of the theory underlying the various topics covered, it would be possible to use individual chapters in a number of different courses. For example, the chapter on non-parametric methods provides an adequate introduction to this area of the subject, while the chapter on the linear model could be included in a third-year course on theoretical statistics.

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Statistical and Mathematical Aspects of Pollution Problems

edited by John W. Pratt

1st edition, 392 pages, Marcel Dekker, New York, 1974, \$19.50

This book consists of twenty five papers, presented in 1973 at an interdisciplinary symposium bringing together environmental scientists and statisticians. It is wisely divided into five sections, not from the viewpoint of the pollution expert, but from that of the statistician. The five sections are:

(i) *Collection and dissemination of basic data.* The four papers deal chiefly with air pollution data. The treatment is purely verbal, no mention being made of any randomised sampling techniques. One paper that claims to optimize the siting of air pollution measurement stations uses a common sense approach, and the method is not shown to be

optimal in any sense. The chief value of the section is that it reveals to the statistician the immense problems associated with the acquisition, verification and handling of the enormous quantities of data generated by these studies.

(ii) *Indices, economic and policy perspectives.* Three papers consider the legal and economic aspects of pollution problems, and the enforceability of the standards laid down by present U.S. pollution control legislation. A further two papers cover the development of environmental indices, similar to the familiar economic indices. The current position, in the view of one of the contributors, is "We need comprehensive and reliable data and facts on which to make decisions. Right now we are in very poor shape as far as this is concerned."

(iii) *Mathematical models.* For the first time we encounter statistical methods. There is a probability model for the detection of stench, a statistical model for the growth reduction of forest trees caused by sulphur dioxide, a model for the mixing of water in estuaries and rivers.

(iv) *Statistical studies using regression and related methods.*

(v) *Other statistical methods.* The regression papers exhibit clearly an understanding of the pitfalls for the unwary. Great care is taken not to fall into the cause-effect trap. Three out of five papers deal with the relationship between air pollution and health. The statistical methods used in the remaining papers include spectral analysis, random sampling (survey into public attitudes), distribution fitting, time series analysis, quantal regression, and a sequential screening programme.

In well chosen closing comments, Gerald van Belle notes: "The statistical community has an obligation to inform consumers of statistical methods about the limitation of the scales, indices and models that are used."

The book provides valuable insights for any statistician becoming involved in environmental problems. But if this is the work of the leaders in the field, as the book claims, then there is clearly great scope for anyone willing to work in a challenging area for statistical applications.

The environmental scientist, with all his pollution monitoring gear, is guilty of generating a new kind of pollution — information pollution. It's over to the statistician to clear up the mess.

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Personelle und Statistische Wahrscheinlichkeit
by Wolfgang Stegmüller

First edition in two volumes. Volume 1: *Personelle Wahrscheinlichkeit und Rationale Entscheidung* xvii + 548 pages. Springer-Verlag. Berlin-Heidelberg-New York, 1973, DM89, US \$31.40.

Volume 2: Statistisches Schlieszen, Statistische Begründung, Statistische Analyse. xv + 419 pages. Springer-Verlag, Berlin-Heidelberg-New York, 1973, DM68, US \$25.20

This is the fourth book in Stegmüller's general series "Probleme und Resultate der Wissenschaftstheorie und Analytischen Philosophie". As the title suggests, this work is generally devoted to an examination of various theories of personal (or subjective) and statistical probability, notably those of R. Carnap, I. Hacking (who appears mysteriously as J. Hacking), C.G. Hempel, R.C. Jeffrey and W. Salmon.

Stegmüller's basic thesis is that the problem of induction should be replaced by two classes of problems: the *practical* and *theoretical* successor problems (Nachfolgerprobleme); and he examines the interdependence and distinction between these two classes, as well as the method of drawing appropriate boundaries between them. An important class of practical successor problems is explored under the heading of *rational decision theory* or the *logic of decision*, in which the concept of personal probability is of supreme importance.

The first part (which is independent of the rest of the book) of the introduction, contains some observations on the problems and objectives of the theory of knowledge. Parts two to four contain fundamental remarks on the themes "probability", "theoretical concepts", and "induction".

Section 0 provides a brief introduction (didactic rather than systematic) to the most important concepts of probability theory and statistics, together with some results in measure and integration theory. Then follows a section based on Jeffrey's work, whose thesis, in Stegmüller's opinion, is, of all attempts, the most interesting and most attractive. One can hardly fail to agree that Jeffrey's "The logic of Decision" is certainly, for one who is neither a logician nor a mathematician, an excellent introduction to the topic.

The second section is based chiefly on some of Carnap's hitherto unpublished writings, and provides a new purely decision-theoretic interpretation of Carnapian proposals. This interpretation disposes of a number of objections to Carnap's theory – for instance, the controversy between Popper and Carnap about induction is seen to resolve itself into a distinction between the theoretical and practical successor problems. Apart from rejecting Carnap's model-theoretic point of view in favour of a linguistic approach, Stegmüller manages to dispense with Carnap's "numerical stenography" – a distinct improvement.

The third section essays a bridge between the philosophico-probabilistic consideration of trials and confirmation of hypotheses on the one hand and special investigations into "statistical inference" (statistischen Schlieszen) – in which the concepts of support and likelihood and concomitant questions are examined – on the other. For different reasons the theories of both Carnap and Popper are inapplicable here, and Stegmüller carries his analysis of "statistical inference" further. Stegmüller's approach draws on Hacking's "Logic of Statistical Inference" (which contains, in Stegmüller's opinion, a somewhat strange mixture of systematic statements and aphoristic remarks), but deviates from the views expressed there in a number of respects – for example, Stegmüller considers Hacking's "law of Likelihood" to be false, and questions the entire likelihood-support theory. This third section, compared to the others, is relatively incomplete, and contains more suppositions, suggestions and critical remarks – albeit categorically garbed

— than systematic analysis.

The fourth section provides a critical examination of two theories of statistical explanation — those of Hempel and Salmon. Stegmüller's substitute thesis can best be rendered in the catchword "statistical proof instead of statistical Hempel-explanation" and "statistical analysis instead of statistical Salmon-explanation".

There are three appendices: (1) Indeterminism of two types (briefly discussed in the first volume). (2) The representation theorem of de Finetti, and (3) Metrization of probability fields.

This weighty two-volume work is a welcome addition to the literature on the foundations of statistics. That it is published in German is of course a disadvantage to the bilingualist (in the South African sense of the word), and one could well look forward to an English translation. The book contributes usefully to current discussions on various "theories of probability", and should find a place on the bookshelves of any mathematical statistician who is interested in the foundations of his discipline.

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Biostatistics in Pharmacology Volume 1 & 2

Edited by A.L. Delaunois

Vol. 1, 640 p.; vol. 2, 488 p., Pergamon Press Ltd, Oxford, 1973, £ 18.00

These two volumes are part of a three-volume series, and they in turn form part of the International Encyclopedia of Pharmacology and Therapeutics which, it is intended, will be made up by approximately 250 volumes. The purpose of the encyclopaedia is said to be "to collect, analyse, assess, integrate and disseminate the available data important to pharmacology and therapeutics ..."

Nine authors have contributed to these volumes, and their style, presentation and choice of content vary from very bad to very good.

First, the very good. Chapter 16 is entitled "*Principles of Biological Assay*". It is rich with examples and covers the whole field of Finney's "Statistical Method in Biological Assay". Presentation is lucid and all necessary formulae are given without any burdensome mathematical details. Unfortunately, the first two pages of this chapter were completely missing from my bound volume.

Next, the very bad. Chapters 7 to 13 are apparently intended to form the theoretical basis for the two volumes. The language is cumbersome and confusing. Apparently through translation, the complete meaning of some sentences is lost. For example, the introductory paragraph under the title "*The method of least squares*", page 621, reads:

"It is important to know that the form of the distribution must be known. In the method of least squares the form need not be known, but the method can also be used if the distribution is known"

Page 708, in the discussion of the relative merits of certain experimental designs reads: "It is easy to apply in a laboratory because of its monotony".

Carelessness appears in some places. For example, on page 619 the same symbol is used for likelihood and log likelihood even in the same formulae, with no explanation at all. Greek and Latin symbols for population parameters and sample estimates are mixed. For example, on page 640 we read "The t distribution is independent of μ and s ".

Errors which might be due to printing are numerous. For example:

page 615: "The standard distribution of the mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} "$$

page 640: When discussing the t distribution, " $E(t) = \sigma$ "

Page 641: When discussing the Chi-square distribution "Its characteristic function is $Q(t) = (1-2it)^{-1/2}$ ".

Not all chapters are as bad as this. Chapter 14, which deals with sequential analysis, is well written and full of well illustrated and useful methods.

Considering the two volumes overall, I definitely could not recommend their purchase. It is, at the same time, disturbing that volumes intended as part of a learned work of reference could be so poorly assembled.

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Examples and Exercises in Operations Research

by A. Ghosal, S.G. Loo and N. Singh

First Edition, 262 pages, Gordon and Breach Science Publishers, Ltd, London, 1975, £ 8.50

This book is intended for use in conjunction with some standard text for a first course in Operations Research. It is based on the dictum that "examples are better than precept" and thus contains numerous applications of practical O.R. In fact, possibly the main feature of the book is its format. The material is presented entirely in the form of questions and answers (or a hint on how to obtain the solution). For example, the chapter on Optimization Theory begins with the question: "Define an optimization problem?", followed by the solution. This is a novel approach to the teaching of O.R. and one which, I feel, has considerable merit.

The book comprises of nine chapters covering a wide range of O.R. topics and including many problems which are bound to interest all readers. There are chapters on model building, the use of statistical methods, optimization theory, inventory control, queueing theory, reliability, combat analysis (interesting in that it is not covered in many books), cybernetics and simulation.

Unfortunately, many of the questions posed in the text are of a theoretical nature, either involving the derivation of various "standard" formulae or are otherwise problems

of a general kind (eg. if x is the demand which has d.f. $F_X(x)$ and C_1 and C_2 are the costs, etc.). If the authors had kept to their dictum and presented only numerical examples, assuming that the theory could be obtained from a standard text, one feels that the book would have far wider appeal. In addition, solutions to many of the more interesting problems are not provided (although a hint is given).

In summary, then, a book with an unusual approach to the subject and many interesting examples with more appeal, perhaps, for a lecturer than a class in general.

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Algebraic Topology, A First Course

by Max K. Agoston

Pure and Applied Mathematics Series, Volume 32, Marcel Dekker Inc., New York and Basel, 376 Pages, illustrated (1976), Price \$23.50

This book is meant for undergraduates. It grew out of notes for a semester course on geometry given by the author at Wesleyan University, Middleton, Connecticut, U.S.A. and at the University of Auckland in New Zealand (the author is now at Stanford University). The structure of the book is very simple indeed. It does not prerequisite any mathematical knowledge at university level but an acquaintance with linear algebra would be helpful. No knowledge of topology whatsoever is presupposed; all is kept in Euclidean n -space.

The subjects treated are: polyhedra, the classification of surfaces, homology groups up to the calculation of the homology groups of the cone, the simplicial approximation theorem. Homotopy is defined but more emphasis is laid on homology theory. The Brouwer-degree is treated in extenso including the Borsuk-Ulam theorem.

Another special feature of the book is the historical discussion given in each chapter. There is also a bibliography of eighteen pages.

In the reviewer's opinion the book can be recommended for an honours course in South Africa, although more homotopy theory would be desirable.

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